Computer Vision:

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Motion in an Image Sequence

- **Motion in an image** sequence refers to the change in position of objects or pixels over time across consecutive video frames. It occurs when an object moves, the camera moves, or both.
- In computer vision, motion is analyzed by **comparing two or more successive frames** to estimate how and where pixels have moved, which helps in tasks such as object tracking, activity recognition, and video surveillance.

Motion in an Image Sequence

For Exampe: If a person walks across a video, their pixel locations change from frame to frame — this change represents motion.

-Key causes of motion:

Object movement

Camera movement

Scene depth variation

Lighting changes

Often uses:

Object tracking

Optical flow estimation

Human action recognition

Video compression

Autonomous driving

Motion Estimation

Finding how much and in which direction objects move between frames.

It estimates:

- Horizontal movement (x-direction)
- Vertical movement (y-direction)

A pixel moves:

Frame 1: (x, y)

Frame 2: (x+2, y+1)

*Motion estimation finds a **motion vector**:*

$$\vec{v} = \begin{pmatrix} \\ dy \end{pmatrix}$$

Types of Motion Estimation

1. Pixel-based Motion Estimation

- Tracks intensity change of each pixel
- Used in optical flow
- Very accurate but computationally expensive

2.Block-based Motion Estimation

- Image divided into blocks (e.g., 16×16)
- Each block searched in next frame
- Widely used in video compression (H.264, HEVC)

3. Feature-based Motion Estimation

- Detects keypoints (corners, edges)
- Tracks features across frames
- Used in object tracking and SLAM

Mathematical Idea

*Assumption of **brightness constancy**:*

$$I(x, y, t) = I(x + dx, y + dy, t + 1)$$

- **Optical flow equation:**

$$I_x u + I_y v + I_t = 0$$

Component	Definition	Mathematical Expression
Horizontal Velocity (u)	Change in x over time	$u = \frac{dx}{dt}$
Vertical Velocity (v)	Change in y over time	$v = \frac{dy}{dt}$
Spatial Gradient (I_x, I_y)	Change in brightness across space	$I_x = \frac{\partial I}{\partial x}, \quad I_y = \frac{\partial I}{\partial y}$
Temporal Gradient (I_t)	Change in brightness across time	$I_t = \frac{\partial I}{\partial t}$

Method	*Description*
Frame Differencing	Simple subtraction between frames
Optical Flow	Pixel-level motion estimation
Block Matching	Motion per block
Lucas–Kanade	Local optical flow
Horn–Schunck	Global optical flow
KLT Tracker	Feature-based tracking
Deep Learning	CNN + FlowNet, RAFT

Optical Flow

The apparent motion of pixels between two consecutive frames.

***-Simply*: Optical flow tells how each pixel moves.**

Optical flow detects that rightward movement:

-When car is at left side (Frame-1)
and slightly move right (Frame-2)

Brightness constancy

Pixel intensity does not change:

-Same pixel, just movement

$$I(x, y, t) = I(x + dx, y + dy, t + 1)$$

Optical Flow Intuition

Imagine: You watch two video frames quickly

Your eyes can tell what moved

Optical flow tries to do the same mathematically.

$$\mathbf{u} : \textbf{Horizontal Velocity}(\textit{the rate of change along the } x-\textit{axis}).$$

$$\mathbf{v} : \textbf{Vertical Velocity}(\textit{the rate of change along the } y-\textit{axis}).$$

Classic Optical Flow Methods:

- Lucas–Kanade Method *(Often applied method)*
- Horn–Schunck Method

Key idea: Instead of using one pixel, use a small window around the pixel. Because one pixel alone is not enough to estimate motion.

$$3\times 3\, or\, 5\times 5\, window$$

Lucas–Kanade assumes:

- motion is same inside a small patch
- pixels in patch move together.

Procedure

I. Take a small window around pixel
II. Compare window in next frame
III. Find displacement that best matches

Feature Tracking Basics:

- Instead of tracking every pixel, we track.

Such as:

I. Corners
II. strong edges

Famous feature detector:

- Harris Corner Detector

Corners are best because:

I. flat area → no information
II. edge → ambiguous motion
III. corner → unique motion

Motion Estimation and Optical Flow

- Motion in image sequences
- Motion estimation concept
- Optical flow intuition
- Lucas–Kanade optical flow
- Feature tracking basics

1 Motion in Image Sequences

- Motion is the **change of object position between consecutive frames** in a video.
- Pixels move as objects move.

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2 Motion Estimation Concept

- Motion estimation **finds how much and in which direction objects move** between frames.
- Represented as **motion vectors (dx, dy)**.

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3 Optical Flow Intuition

- Optical flow tells **how each pixel moves between frames**.
- It’s like tracking **pixel arrows showing direction and speed**.

4 Lucas–Kanade Optical Flow

- Estimates optical flow **using a small patch of pixels** instead of single pixels.
- Assumes **all pixels in the patch move together**.

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5 Feature Tracking Basics

- Track only **good features (corners or textured points)**, not every pixel.
- Good features are stable and easy to follow across frames.

- To analyze the movement of this pixel, we calculate the **displacement vector**, often referred to in computer vision as the **optical flow**.

$$A\ pixel\ moves\ from\ position$$

$$\left(x,\, y\right) \;=\; \left(10,\, 15\right)\ in\ frame\ 1\ to\ \left(x,\, y\right) \;=\; \left(13,\, 18\right)\ in\ frame\ 2.$$

$$Find\ \ the\ motion\ vector\ \left(dx,\, dy\right).$$

$$Find\ \ the\ magnitude\ of\ \ motion.$$

$$dx = 13 - 10 = 3 \; dy = 18 - 15 = 3$$

$$Motion\ vector = (3,\ 3)$$

$$Magnitude = \sqrt{}(dx^2 + dy^2)$$

$$= \sqrt{}(3^2 + 3^2) = \sqrt{}18 \approx 4.24$$